WORKSHEET

Chapter 9: Sampling methods

1 State the kind of sampling used in each of the following non-random samples of students at a school.

a Max placed the names of students in his class into a hat and drew out six names.

b Nerida chose the closest six people to her.

c Jamal went through the class list and selected every sixth student in order.

d Heidi divided the class into six groups of three students each. She then randomly chose two groups.

e Emma divided the class into two equal age groups. She then randomly chose three students from each group.

f Jacob asked six people to volunteer.

g Olivia chose the six tallest people in the class.

h Jayden selected students until he had selected two boys and four girls.

i Abby numbered the students in her class from 1 to 28. She then used a random number table to select six numbers between 1 and 28 (inclusive). She chose the students with assigned numbers that matched the random numbers.

j Ethan divided the class into groups of students who live in the same suburb. He then randomly chose two from each group.

k Helena chose the six girls she knew to be the best netball players.

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2 Consider the population of all dingoes in central Queensland. The sample of that population that farmers observe is largely the dingoes that prefer to live near a property. They also like to eat farm animals.

The farmers concluded that all dingoes are dangerous and got the government to assist in distributing a poison bait to reduce the overall dingo population. The overall population of dingoes was reduced, but the farmers still lost almost as many sheep as before.

a Is the sample the farmers observed a random sample?

b If not, is it safe to use the results to describe the entire population? Why?

c Is the idea of reducing the size of the entire population justified based on the farmers' experience?

3 Do the following procedures give a representative sample for the entire population of Melbourne? Why or why not?

a Select every third woman entering a particular hairdressing salon.

b Select every third person coming out of a football match at the MCG.

4 In an online survey about computer use, people who visit a particular site are invited to complete a survey containing a series of questions about how they use computers. The results of the first 20 respondents are used.

a What type of sampling is this?

b Identify any sources of bias likely in this survey.

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5 A call-in radio program asks listeners a question regarding gun control. The response of every third caller is used.

a What type of sampling is this?

b Identify any sources of bias likely in this survey.

6 A researcher is sent into the city centre and asked to interview people who earn high incomes (>$150 000 p.a.) regarding a new tax being considered by the government. The researcher found that people were reluctant to reveal their income so she chose interviewees she considered to be dressed in expensive clothes as she believed that these people would be earning a lot of money.

a What type of sampling is this?

b Identify any sources of bias likely in this survey.

7 A researcher is employed to sample the daily takings of a business that operates every day of the week. He decides to select every seventh entry in a list of daily takings.

a What type of sampling is this?

b Identify any sources of bias likely in this survey.

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Answers

1 **a** Simple random sampling

b Convenience sampling

c Systemic sampling

d Cluster sampling

e Stratified random sampling

f Convenience sampling

g Judgement sampling

h Quota sampling

i Simple random sampling

j Stratified random sampling

k Purposive sampling

2 **a** No

b No. The results are biased in that they are unrepresentative of the population.

c No

3 **a** Not representative. Many examples of bias including: only women are selected, only people from a particular location are selected, only people who use hairdressing salons are selected, etc.

b Not representative. Many examples of bias including: only people who like football are selected, only people who can afford to attend the MCG are selected, only people who are able to attend the MCG are selected, etc.

4 **a** This is an example of convenience sampling.

b There are many sources of bias including: only people who are online are selected, only people who visit a particular site are selected, only people who are more interested in technology than is typical are selected, etc.

5 **a** This is an example of systemic sampling.

b Many sources of bias exist including: only people listening to the radio program are selected, only people with strong views on the subject are selected, etc. The resulting sample tends to over-represent individuals who have strong opinions.

6 **a** Judgement sampling

b There are many sources of bias including: people who dress in expensive clothes may not earn high incomes, people who dress in expensive clothes may not be working at all, etc.

7 **a** Systemic sampling

b Bias exists because the sample would be comprised of observations that were all on the same day of the week.

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